

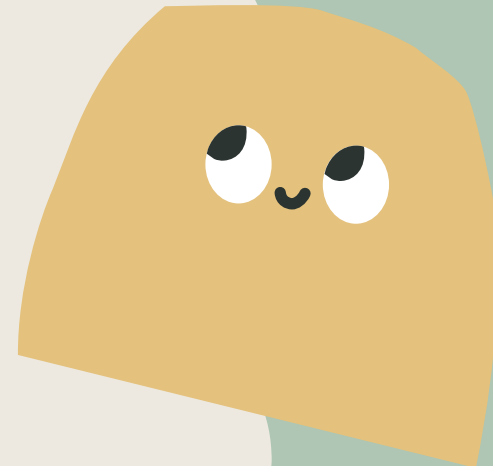
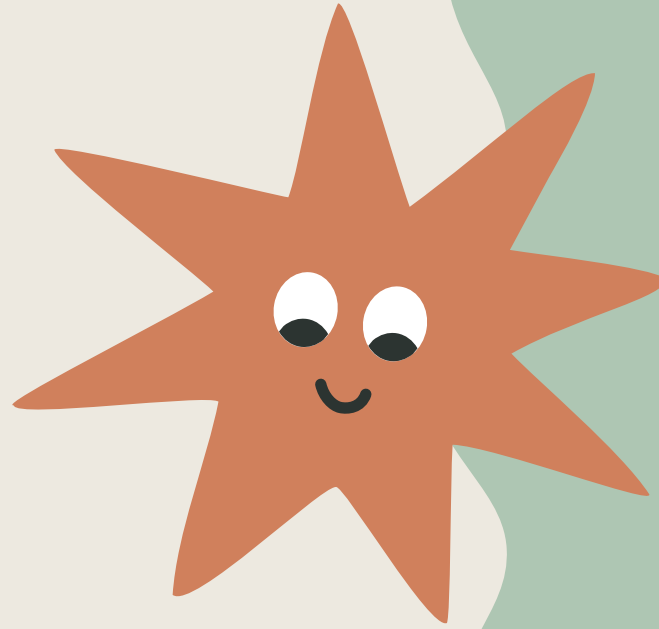
UNIT 1

FUNDAMENTALS OF

COMPUTER

9TH COMPUTER SCIENCE
SINDH TEXTBOOK BOARD

AWINASH GOSWAMI



Agenda

Evolution of Computers

Classification of Computers

Role of Computer

Careers in IT Field

Computer Hardware

Basic Operations of A Computer

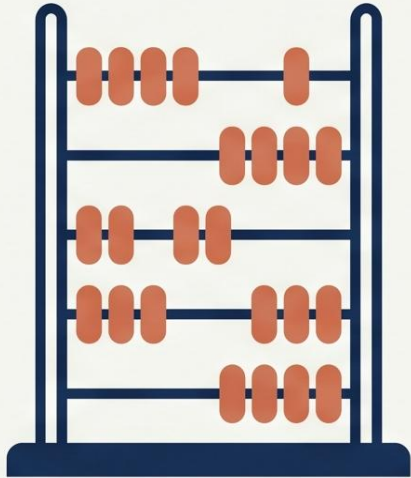
Computer Software

Introduction to computer

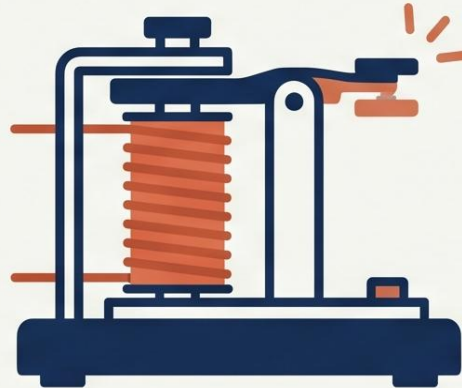
A computer is an electronic data processing machine.

EVOLUTION OF COMPUTERS

MECHANICAL ERA
(DARK AGE)



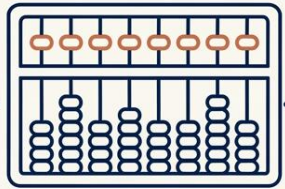
ELECTRO-MECHANICAL
ERA (MIDDLE AGE)



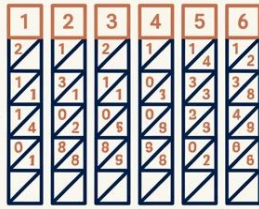
ELECTRONIC ERA
(MODERN AGE)



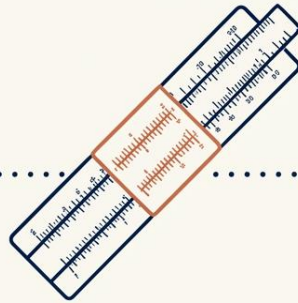
MECHANICAL ERA (DARK AGE)



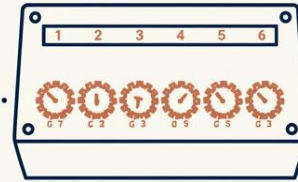
Abacus
(3000 B.C.)



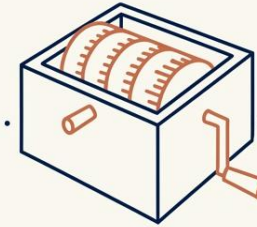
Napier's Bones
(1612 A.D.)



Slide Rule
(1622 A.D.)



Pascaline
(1642 A.D.)



Leibniz's Calculator
(1694 A.D.)



Babbage's Engine
(1822 & 1837 A.D.)

Abacus (3000 B.C.)

- Abacus was invented about 5000 years ago.
- It is also known as counting frame.
- Abacus is still used to teach basic arithmetic operations to the students.
- Abacus is considered as first computer prototype.



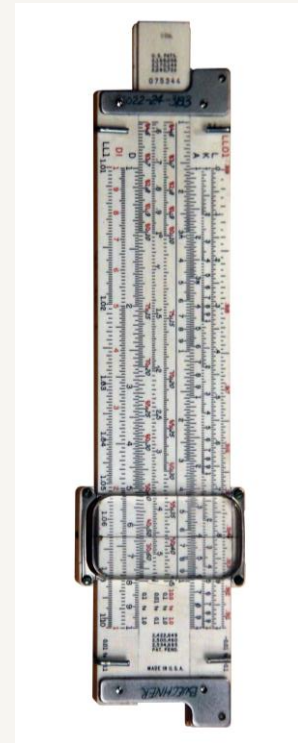
Napier's Bones (1612 A.D.)

- Scottish mathematician John Napier developed Napier's Bones, in 1612.
- It is also called Napier's Rods.
- It was a small machine that contained ten rods. The rods were made up of bones, wood or metal.
- Napier's Bones was used for multiplication and division.



Slide Rule (1622 A.D.)

- English mathematician William Oughtred invented a Slide Rule.
- This was basically a pair of two moveable rules placed side by side.
- They had a standardized set of markings called scales. These scales were used for mathematical computations.
- Slide Rule was used to multiply and divide numbers. The modern slide rules are still used in some countries.



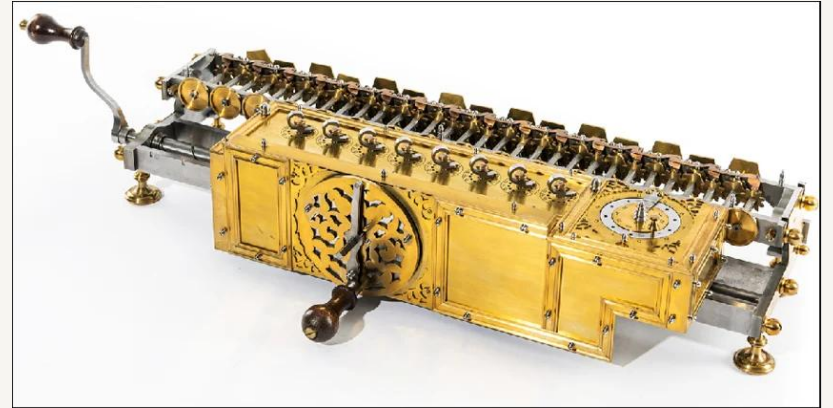
Pascaline or Pascal's Calculator (1642 A.D.)

- French mathematician Blaise Pascal invented a mechanical calculator which was called Pascaline.
- He developed this machine to help his father who was a tax collector.
- Pascaline can perform addition and subtraction by eight figures. User had to adjust wheels in order to perform these calculations.



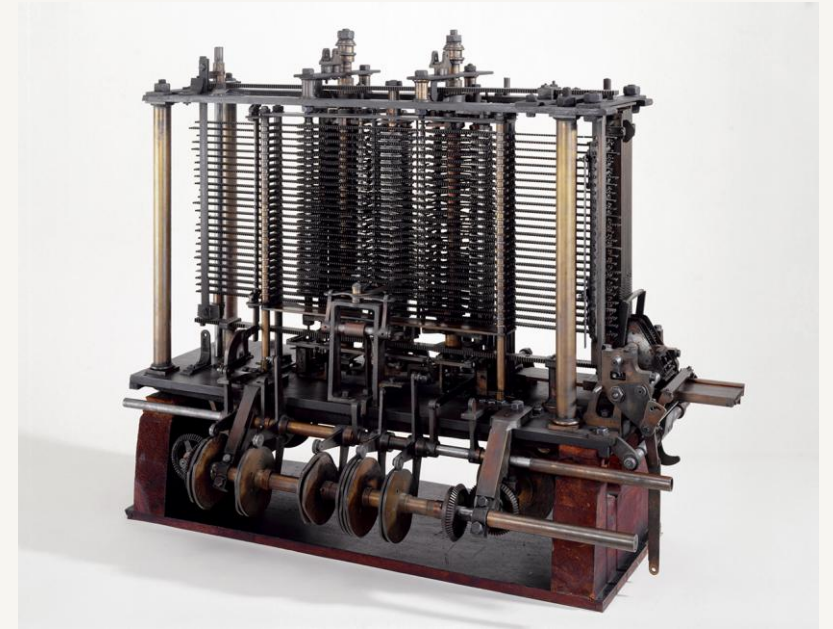
Leibniz's Calculator (1694 A.D.)

- Leibniz's Calculator was developed by the German mathematician Gottfried Wilhelm Leibniz.
- It was also called stepped reckoner.
- It was the first calculator that could perform all four basic arithmetic operations; addition, subtraction, multiplication and division



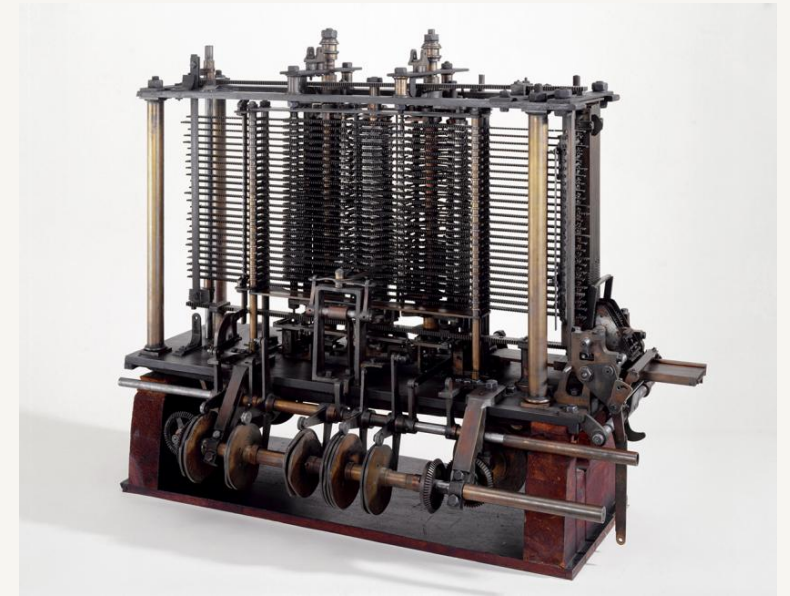
Babbage's Difference and Analytical Engines (1822 and 1837 A.D.)

- Charles Babbage, an English mathematician and engineer, is called the Father of Computers for creating the first complete computing machine.
- His first invention, the Difference Engine (powered by steam), was a large metal calculator that:
 - *Stored data temporarily in mechanical memory.*
 - *Printed results from calculations using difference equations.*

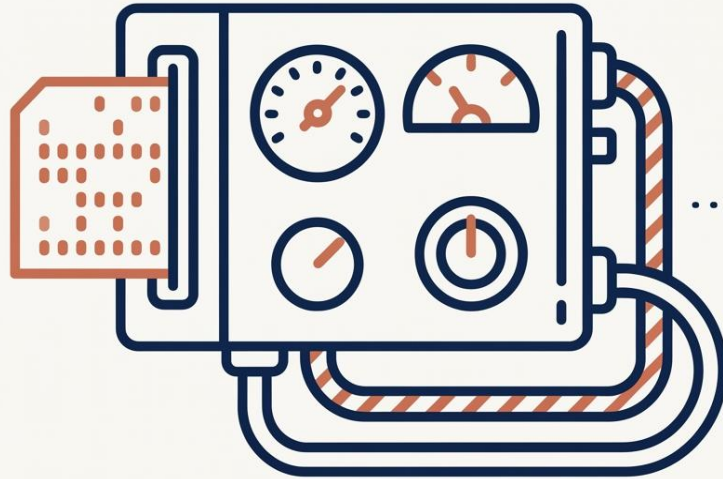


Babbage's Difference Cont....

- In 1837, he designed the Analytical Engine, which he never built. Its advanced design included:
 - *An ALU (Arithmetic Logic Unit).*
 - *Programmatic flow control.*
 - *Punched-card programming and integrated memory.*
- Historians view it as the first concept of a general-purpose computer due to its comprehensive design.



ELECTRO-MECHANICAL ERA (MIDDLE AGE)



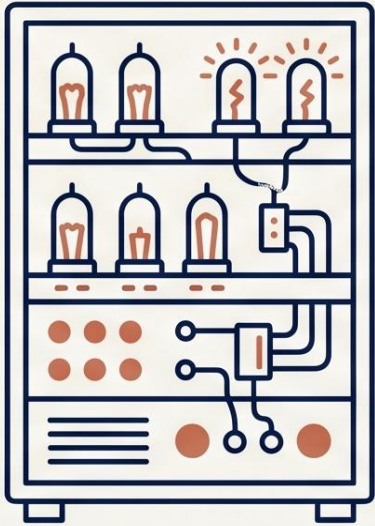
Hollerith's Tabulating Machine (1890 A.D.)

Herman Hollerith's Tabulating Machine (1890 A.D.)

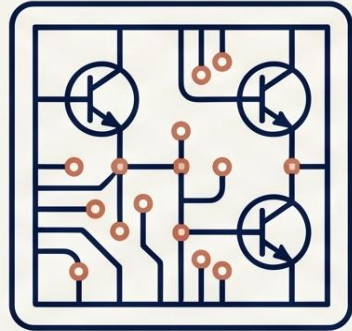
- In 1890, American inventor Herman Hollerith invented the Tabulating Machine. It was first used for the U.S. Census that year.
- While the 1880 census took 7 years to complete, Hollerith's machine finished the 1890 census in just 6 months—a huge speed boost.
- The machine became very popular and was widely used in many U.S. government offices.



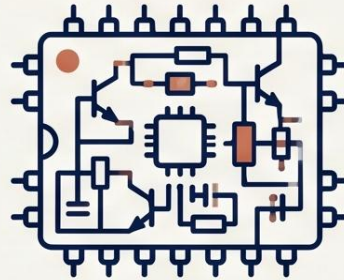
ELECTRONIC ERA (MODERN AGE)



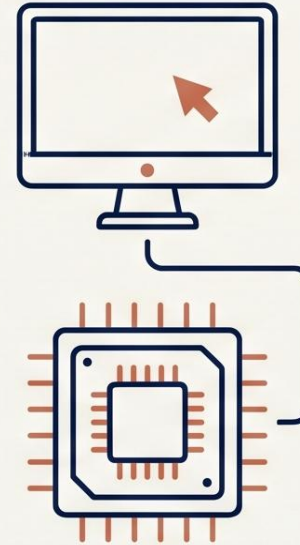
First Generation
(1940–1956)



Second Generation
(1956–1963)



Third Generation
(1964–1971)



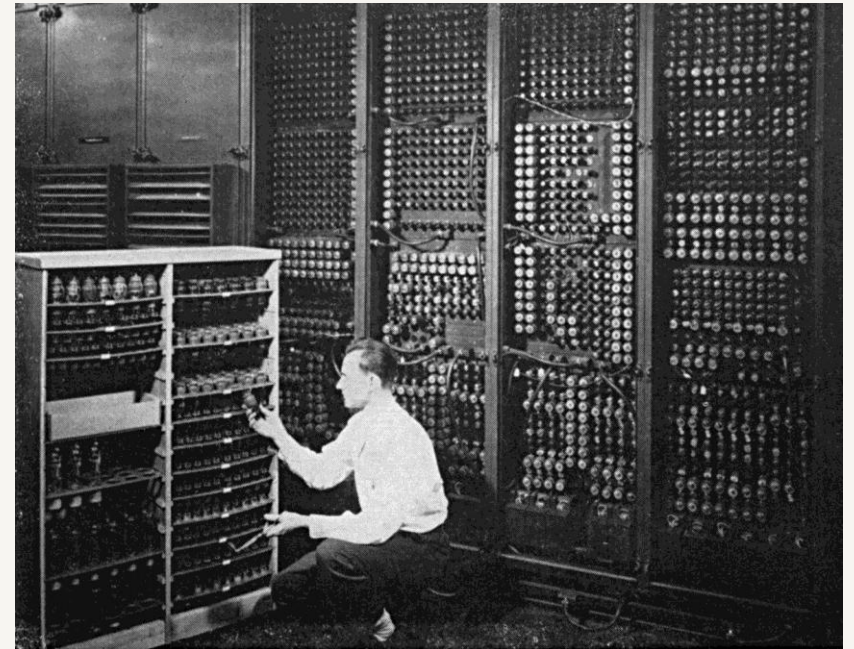
Fourth Generation
(1971–Present)



Fifth Generation
(Present and Beyond)

First Generation of Computers (1940 to 1956)

- First-generation computers (based on vacuum tubes) were very large in size. They used:
 - *Machine language (1s and 0s) for programming.*
 - *Magnetic drums as primary internal storage.*
 - *Punched cards for input.*
 - *Batch processing operating systems.*
- These computers were mainly used for scientific and research purposes. Examples are ENIAC and UNIVAC.



Second Generation of Computers (1956 to 1963)

- Second-generation computers used transistors, making them smaller, faster, cheaper, and more efficient than first-generation machines.
- Key features:
 - *Assembly language and high-level language FORTRAN were introduced.*
 - *Magnetic core replaced drums as primary internal storage.*
 - *Punched cards continued for input.*
 - *Batch processing and multiprogramming operating systems were used.*
- These computers were used for commercial production, scientific, and engineering work. Examples: IBM 7094 and IBM 1401.



Third Generation of Computers (1964 to 1971)

- Third-generation computers used Integrated Circuits (ICs), which further decreased size and increased speed and efficiency while making computers less expensive.
- Key features:
 - *High-level languages like Pascal and COBOL were used.*
 - *Keyboard (input) and monitor (output) made computers easier to use.*
 - *Time-sharing and real-time operating systems were introduced.*
 - *Computers were now used for database management and automatic industrial control.*
- Examples: IBM 360 and IBM 370.



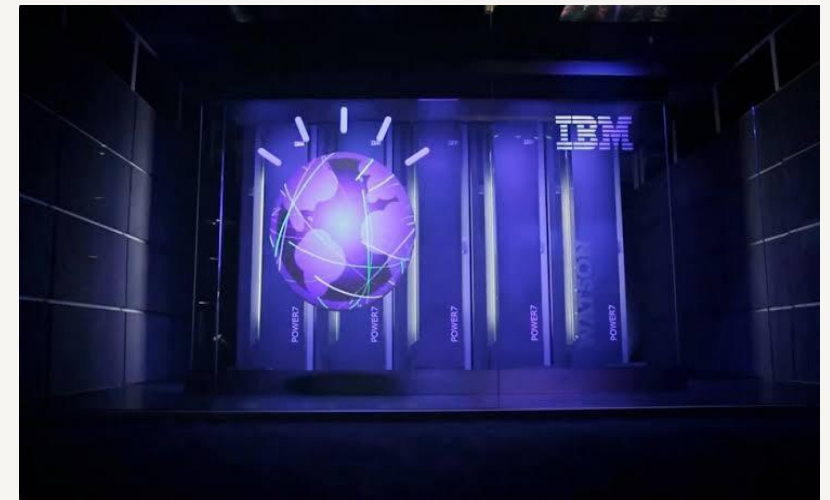
Fourth Generation of Computers (1971 to Present)

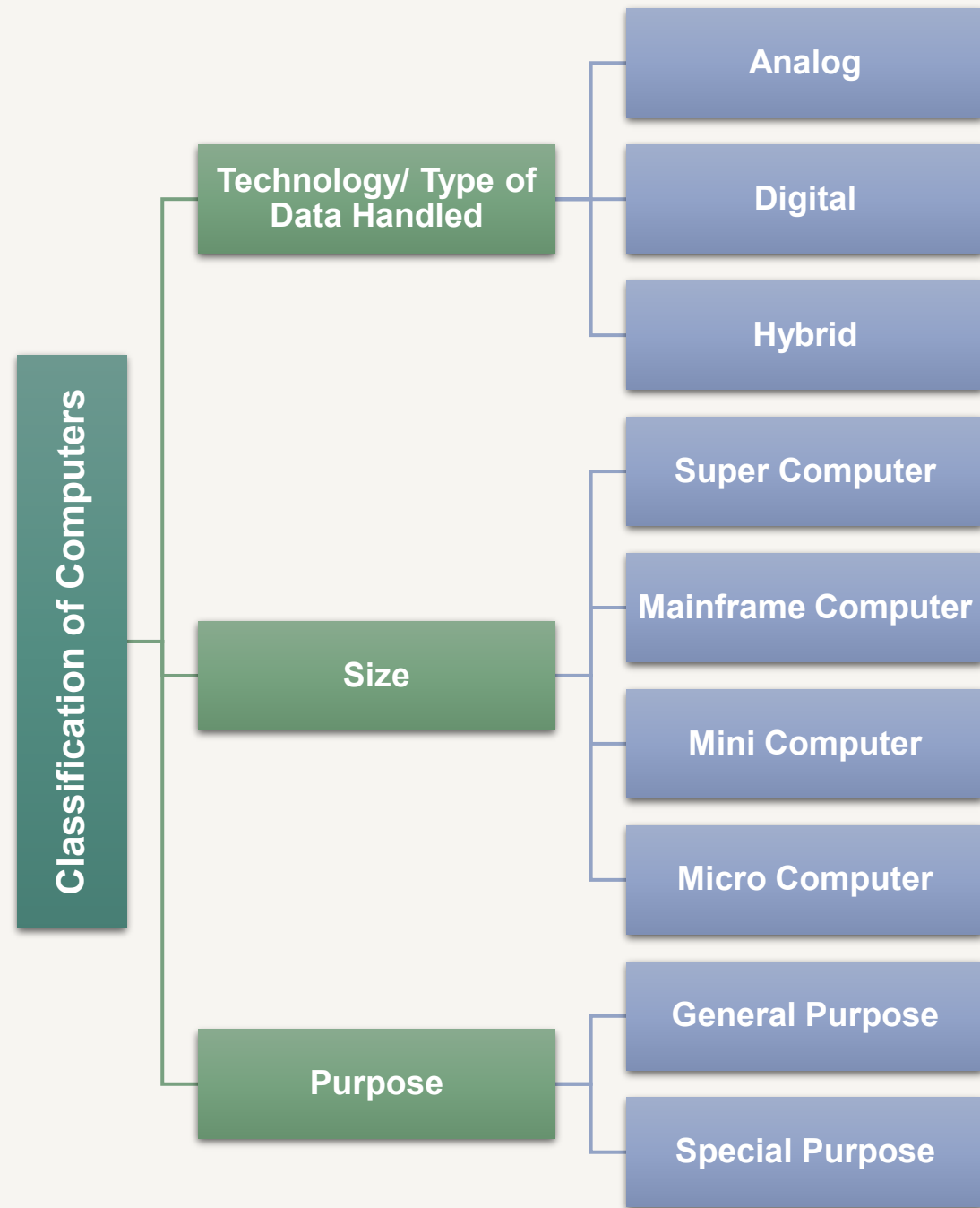
- Fourth-generation computers used microprocessors, which was revolutionary. They became faster, cheaper, smaller, and more reliable.
- Key features:
 - *Used semiconductor memories (RAM and ROM); magnetic storage became popular.*
 - *More high-level languages introduced – C, C++, Java, etc.*
 - *Used in almost every field – space, business, art, and more.*
 - *Operating systems: time-sharing, real-time, and distributed.*
 - *Saw the development of Graphical User Interfaces (GUIs) for easier use.*
- Examples: Apple Macintosh, IBM PC.



Fifth Generation of Computers (Present and Beyond)

- Fifth-generation computers are still being developed. They aim to make computers capable of self-learning, reasoning, and generalization.
- Key features:
 - *Will be able to process human languages and control machines like robots.*
 - *Based on Artificial Intelligence (AI) , which includes branches like Machine Learning, Deep Learning, Natural Language Processing, Robotics, Expert Systems*
- These computers represent the future of intelligent and human-like computing.





TECHNOLOGY BASED



Analog Computers

- Analog Computers are used to process analog data.
- Analog data are in the form of continuously varying physical quantities like pressure, temperature, voltage, speed and weight.
- Examples of Analog computer are speedometer of a car, voltmeter etc.



Digital Computers

- Digital Computers are most commonly used type of computers.
- They are used to process information with quantities using the binary number system (0's and 1's).
- Digital Computers are used in home, educational institutes, offices, business, scientific fields, etc.



Hybrid Computers

- Hybrid Computers are the combination of Analog and Digital Computer system.
- These computers combine analog and digital features of computers in a single machine.
- A Hybrid Computer uses analog to digital and digital to analog conversion. It may input or output either digital or analog data.

SIZE BASED

Super Computers

- Super Computers are the most powerful, fastest and largest computers.
- They are extremely expensive.
- These computers are widely used in scientific applications such as aerodynamics, design simulations, processing of geological data, weather forecasting and nuclear research.

Mainframe Computers

- Mainframe Computers are powerful multi-user and multiprocessors computers.
- They can process huge number of calculations at very high speed.
- Mainframes are also very expensive and require a lot of technical expertise to be installed and operated.
- They are used in banks and many large business organizations where several users work simultaneously.

Minicomputer Computers

- These are smaller than mainframe computers, but they are more powerful than Microcomputers.
- Minicomputers usually use multi-user operating system.
- Multiple users can use the Minicomputers through terminals.
- Minicomputers may be used as network servers and Internet servers.
- DEC VAX and IBM AS/400 are good examples of minicomputers.

Microcomputers Computers

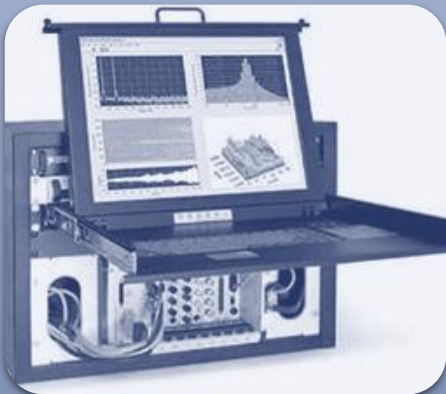
- Microcomputers are also called Personal Computers (PCs).
- The use of microprocessor has made computers cheaper yet faster and more reliable.
- These are the smallest computers designed to be used by individuals.
- PCs can be used for variety of tasks like documentation, calculations, illustration and entertainment.

PURPOSE BASED



General Purpose Computers

Most computers in use today are General Purpose Computers which can process variety of tasks like word processing (typing & editing), payrolls, accounts, inventory control, manipulating facts in a database, making scientific calculations and statistical data analysis, and controlling security system of an organizations etc. Desktops, laptops, tablets and smart phones are examples of General-Purpose Computers.



Special Purpose Computers

As the name states, Special Purpose Computers are designed to perform specific tasks hence they repeatedly perform single job more efficiently. They are also known as Dedicated Computers. These computers are useful in traffic lights control system, navigational system, aviation, weather forecasting, satellite tracking and ATMs.

ROLE OF COMPUTER



Artificial Intelligence (AI)

Artificial Intelligence means simulating human intelligence in machines so they can think and act like humans. Machine Learning is also a type of AI which learns patterns from data.

AI is widely used in scientific experiments, healthcare, and space technologies



Robots

Robots are machines controlled by a computer that performs complex actions. Robotics deals with designing, creating, and programming robots. Robots are commonly used in manufacturing, industry, science, medicine, and education.



Wireless Communication

Wireless communication is used in transmitting information using electromagnetic waves like Infrared, Radio Frequency, and satellite. GPS is used to find real-time locations. Smartphones with 3G, 4G, and 5G offer better coverage and fast mobile internet.



Virtual Reality

Virtual Reality (VR) is an artificial environment created with software. It is presented to the user in a way that makes them feel it is real. It is used for simulating real environments for training and education, as well as for creating imagined environments for games or interactive stories

Careers In IT Field

Career	Role / Description
Software Engineer	Uses programming languages to develop software like games, LMS, business, educational, and entertainment software.
Network Administrator	Manages an organization's network – installs, maintains, and upgrades hardware/software for efficient networking.
Database Administrator (DBA)	Maintains a secure database environment in an office, business, or organization.
Web Designer	Builds websites with attractive designs, images, and text. Responsible for how a site looks, works, and stays updated.
Graphic Designer	Creates layouts for brochures, magazines, ads, and documents using graphics software – combines art and technology.
Information Security Analyst	Protects networks and prevents data loss. Called the "police force of the internet" – stops hackers from stealing account details.
Computer Science / IT Teacher	Teaches computers in schools, colleges, and universities – shows students the importance and impact of computers.

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graph TD; A[Computer Hardware] --> B[System Unit]; A --> C[Input Devices]; A --> D[Output Devices]
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Computer
Hardware

System
Unit

Input
Devices

Output
Devices

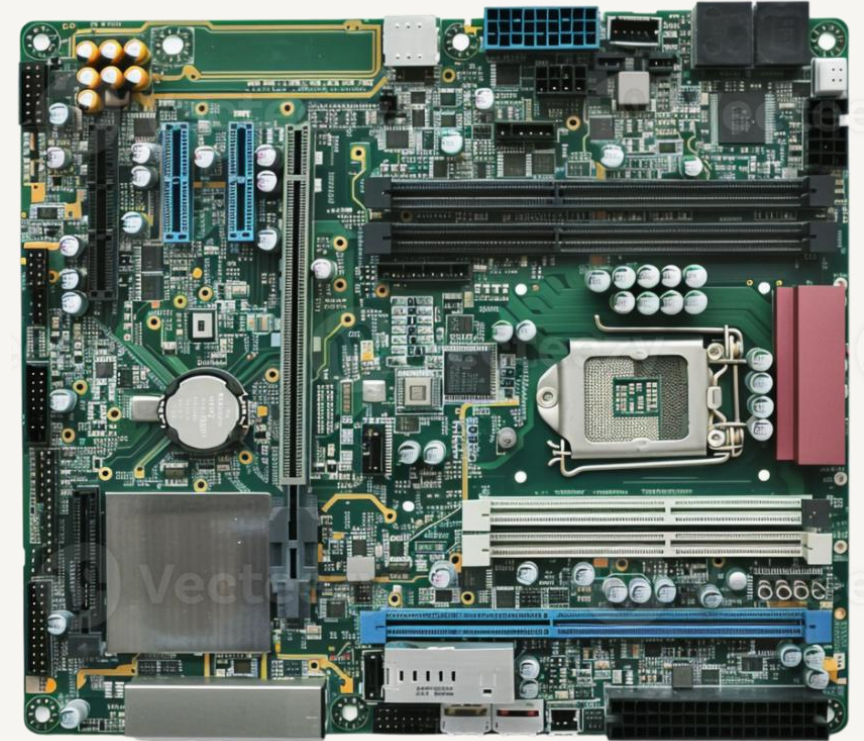
Computer Hardware: System Unit

- A System Unit is the part of a computer that contains the primary devices. The system unit includes the motherboard, CPU, RAM and other components, as well as the case in which these devices are enclosed.



System Unit: Motherboard

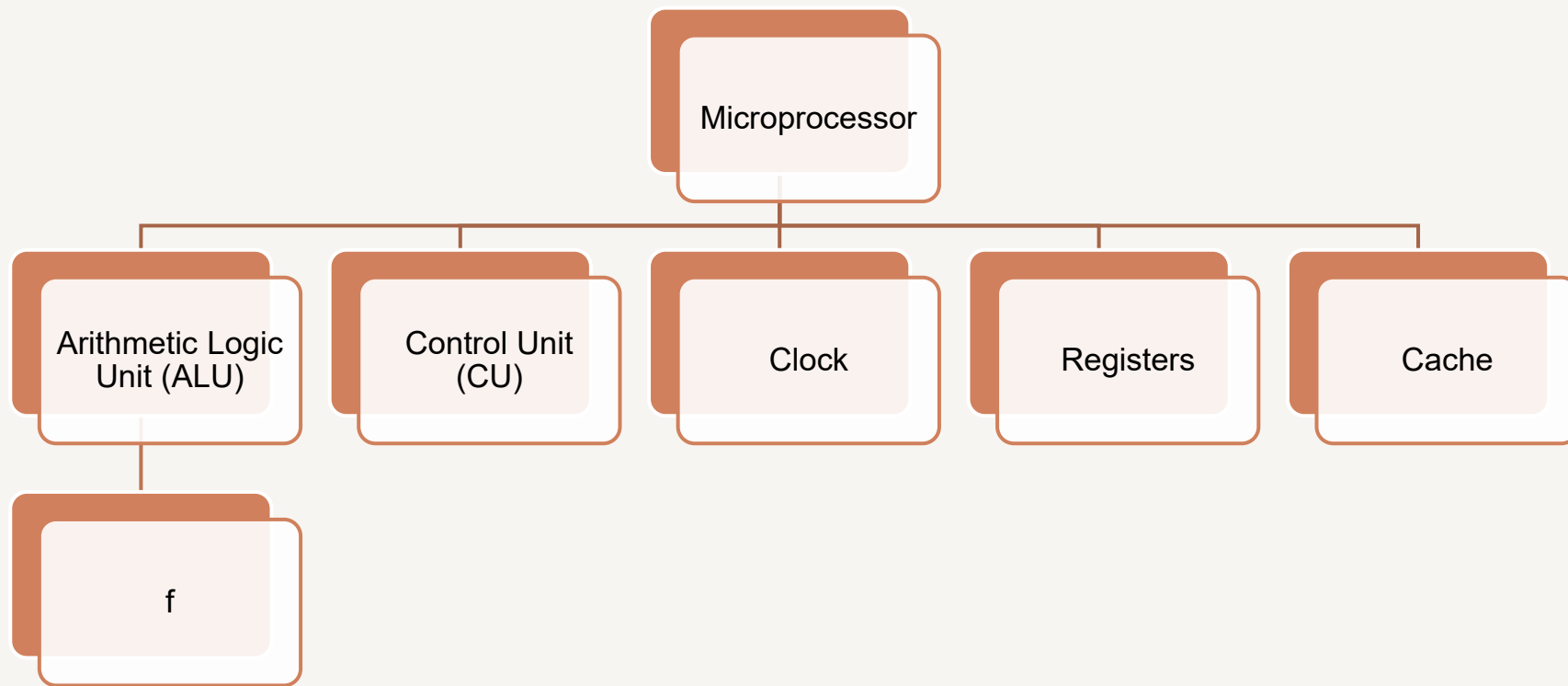
- The motherboard is the main board which connects different parts of computer.
- It includes the following general components: Microprocessor (CPU), Slots, Ports, Buses, RAM, ROM and other electronic components for example resistors, capacitors, diodes, transistors, jumpers etc.
- Just like nervous system it allows communication between all parts of the computer.
- We can find CPU, memory slots, expansion slots and a number of chipsets on motherboard.
- Various components of computer may be attached, detached and upgraded on motherboard according to need of the user. Motherboards also have connectors called ports. These ports are used to connect input, output and other peripheral devices.



System Unit: Microprocessor (CPU)

- CPU or microprocessor is the brain of computer.
- The microprocessor is a chip containing millions of tiny transistors. These transistors manipulate data.
- Microprocessor performs all the calculations necessary to make the computer work. These calculations are performed at very high speed and accuracy.
- Microprocessor is made up of silicon. The microprocessor fetches, decodes, executes and stores all the instructions given by the user or any other device. The speed of computer mainly depends upon the speed of microprocessor. A microprocessor that has faster clock speed (measured in GHz) and more cores and cache works faster. There are typically five components of a microprocessor: Arithmetic Logic Unit (ALU), Control Unit (CU), Clock, Registers, Cache.





BASIC OPERATIONS OF A COMPUTER

- To be used...

COMPUTER SOFTWARE

- To be used...